

ECHS FRAUD ANALYTICS

Geo-Spatial Fraud Clustering Analysis

MODULE 9 REPORT — FULL DATABASE COVERAGE | FY 2013 – FY 2026

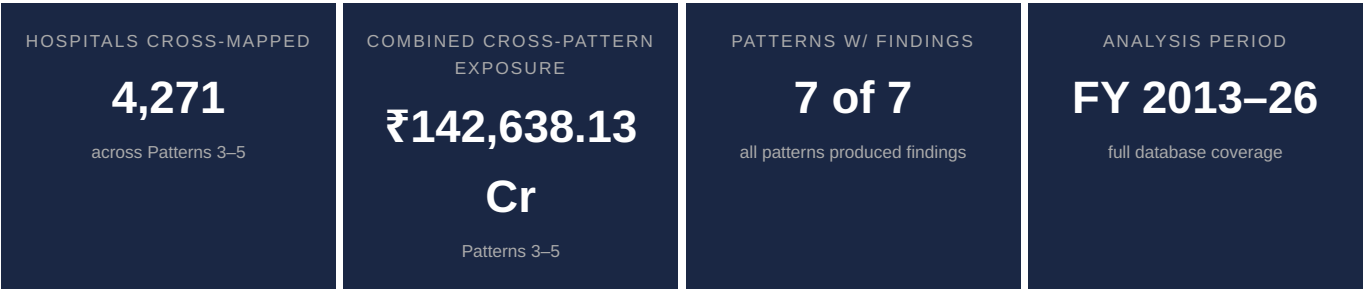
CLASSIFICATION	PERIOD	HOSPITALS SCANNED	PATTERNS RUN	PEAK DEDUCTION
RESTRICTED	FY 2013–26	4,271	7 Patterns	19.13%

IIT KANPUR — Data Analytics & Fraud Intelligence Division

29 June 2026 | Ex-Servicemen Contributory Health Scheme (ECHS)

EXECUTIVE SUMMARY

Module 9 analyses the **geographic and temporal concentration of ECHS fraud**: which regions and hospitals are fraud hotspots, how referral usage crosses regional lines, and which hospitals are emerging as new clusters before they become entrenched. Patterns 1–2 establish the region- and hospital-level deduction baseline (settlement_stat, full database coverage, FY 2013–FY 2026). Patterns 3–7 build on fraud signals already established elsewhere in the ECHS fraud-analytics programme — hospital risk-scoring, referral-syndicate detection, and billing-trend analysis — re-mapping them geographically and temporally without running any new fraud-detection queries.



SEVEN GEO-SPATIAL FRAUD PATTERNS — SUMMARY

#	Pattern	What It Detects	Coverage
1	Regional Fraud Hotspots	Deduction rate by ECHS region, settlement_stat	All 31 regions
2	Top Hospitals by Deduction	Highest absolute deduction hospitals + chain detection	Top 100 hospitals
3	Cross-Pattern Hospital Concentration	Hospitals flagged by the most independent fraud modules	4,271 hospitals mapped
4	Cross-Pattern Regional & State Clustering	Geographic rollup of all already-flagged fraud cases	33 regions, 35 states
5	Cross-Pattern Hospital Chain Detection	Multi-unit hospital brands flagged across patterns	563 multi-unit chains
6	Cross-Region Referral Usage	Polyclinics/entities funnelling referrals to one hospital outside their own city	78 funnelling pairs
7	Predictive — Emerging Fraud Clusters	Hospitals with the sharpest year-on-year billing acceleration	1,066 hospital-year records

IMMEDIATE RECOMMENDED ACTIONS

1. **Joint audit of cross-pattern top-20 hospitals** (4,271 hospitals mapped) — these hospitals are corroborated by multiple independent statistical methods, not a single pattern's threshold; prioritise above single-pattern outliers.
2. **Field audit deployment to R-6 (CHENNAI) and R-1 (NEW DELHI)** — the two regions topping the baseline deduction-rate and cross-pattern exposure rankings respectively.
3. **Corporate-group-level audit of multi-unit chains**, starting with METRO HEART (9 units) and the Park Hospital units — audit the legal-entity group as a whole, not branch-by-branch.
4. **Recovery review for state-level concentration** — Haryana accounts for the largest share of combined cross-pattern flagged exposure (₹37,544.45 Cr) and should anchor the next regional audit cycle.
5. **Audit the cross-region referral pairs in Table 6.1**, starting with VidyaDhar Nagar (VidyaDhar Nagar) → MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED (JAIPUR) — a single referral source sending the overwhelming majority of its patients to one out-of-area hospital is a classic referral-syndicate signature.
6. **Pre-emptive review of emerging-cluster hospitals (Table 7.1)**, led by VIJAY HOSPITAL — these hospitals are still ramping up, so intervening now is cheaper than after the pattern becomes structural like Park Hospital or Vijay Hospital.

7. Extend geo-mapping to the remaining fraud-detection analyses once their exports carry a numeric office hospital identifier — a handful of analyses (identity/duplicate-claims, length-of-stay, diagnostic overutilisation, network collusion) cannot yet be geo-mapped because their existing outputs either have no hospital-level rollup or use free-text identifiers incompatible with the office master's numeric ID space, the same ID-mismatch issue this report's predecessor flagged.

Methodology Note

Patterns 3–7 do not run new fraud-detection queries. They reuse hospital-level summary data already produced elsewhere in the ECHS fraud-analytics programme and aggregate it by hospital office ID. The combined "exposure" figures sum each underlying pattern's own claimed/deducted amount and are a risk-concentration score, not a deduplicated unique-loss total — the same claim may legitimately appear in more than one pattern's denominator.

Pattern 7's year-on-year figures are filtered to hospitals with at least ₹1 Cr of current-year exposure and a meaningful prior-year claim base, to exclude statistically meaningless percentage spikes from near-zero starting volumes.

PATTERN 1

Regional Fraud Hotspot Analysis
settlement_stat — all 31 ECHS regions

REGIONAL FRAUD HOTSPOTS — DEDUCTION RATES BY REGION

Description: Aggregates every claim in settlement_stat by ECHS region code (SS_REGION_ID), computing total claimed, approved, deducted, and deduction rate per region — full database coverage, no date filter. Regions with high deduction rates represent geographic concentrations of overbilling, either from a few dominant high-deduction hospitals or widespread billing manipulation across many facilities. ECHS's own region master names only 13 of the 31 region codes (1–13); the remainder exist in the claims data only as numeric codes with no assigned name, so they are shown as a code alone (e.g. R-20) below.

Table 1.1 — All 31 ECHS Regions, Sorted by Deduction Rate

Region	Hospitals	Claims	Claimed	Approved	Deducted	Ded %
R-6 (CHENNAI)	119	388,080	₹1129.36 Cr	₹913.28 Cr	₹216.08 Cr	19.13%
R-13 (AHMEDABAD)	79	456,933	₹647.04 Cr	₹544.93 Cr	₹102.10 Cr	15.78%
R-3 (KOLKATA)	223	1,000,068	₹1470.32 Cr	₹1243.03 Cr	₹227.29 Cr	15.46%
R-25	83	486,354	₹568.75 Cr	₹481.67 Cr	₹87.09 Cr	15.31%
R-8 (JAIPUR)	266	2,112,770	₹4126.55 Cr	₹3504.28 Cr	₹622.28 Cr	15.08%
R-20	72	105,808	₹171.70 Cr	₹147.57 Cr	₹24.12 Cr	14.05%
R-26	65	287,953	₹336.40 Cr	₹289.63 Cr	₹46.77 Cr	13.90%
R-11 (ALLAHABAD)	61	276,074	₹266.54 Cr	₹232.11 Cr	₹34.43 Cr	12.92%
R-18	147	437,590	₹528.32 Cr	₹460.42 Cr	₹67.90 Cr	12.85%
R-5 (HYDERABAD)	140	858,789	₹1255.36 Cr	₹1094.36 Cr	₹161.00 Cr	12.83%
R-27	209	1,547,779	₹3334.93 Cr	₹2914.05 Cr	₹420.89 Cr	12.62%
R-16	73	142,008	₹204.57 Cr	₹179.79 Cr	₹24.79 Cr	12.12%
R-10 (CHANDIGARH)	86	395,251	₹921.23 Cr	₹812.33 Cr	₹108.90 Cr	11.82%
R-17	141	878,432	₹1221.40 Cr	₹1083.39 Cr	₹138.01 Cr	11.30%
R-29	363	3,285,282	₹4429.62 Cr	₹3932.75 Cr	₹496.87 Cr	11.22%
R-1 (NEW DELHI)	429	4,277,447	₹7904.82 Cr	₹7064.28 Cr	₹840.54 Cr	10.63%
R-24	216	1,065,357	₹2752.50 Cr	₹2465.30 Cr	₹287.20 Cr	10.43%
R-32	30	291,094	₹375.30 Cr	₹339.09 Cr	₹36.21 Cr	9.65%
R-30	161	1,420,453	₹2062.89 Cr	₹1870.08 Cr	₹192.82 Cr	9.35%
R-14	57	240,374	₹274.21 Cr	₹249.58 Cr	₹24.63 Cr	8.98%
R-21	74	443,780	₹551.55 Cr	₹502.24 Cr	₹49.31 Cr	8.94%
R-19	124	563,384	₹688.31 Cr	₹627.63 Cr	₹60.68 Cr	8.82%
R-31	34	210,025	₹173.22 Cr	₹158.49 Cr	₹14.73 Cr	8.51%
R-28	84	467,228	₹763.74 Cr	₹699.81 Cr	₹63.93 Cr	8.37%
R-15	94	740,478	₹1007.49 Cr	₹923.44 Cr	₹84.05 Cr	8.34%

Region	Hospitals	Claims	Claimed	Approved	Deducted	Ded %
R-2 (MUMBAI)	116	3,077,053	₹1911.24 Cr	₹1757.22 Cr	₹154.02 Cr	8.06%
R-4 (BANGALORE)	200	2,642,599	₹5917.03 Cr	₹5490.02 Cr	₹427.01 Cr	7.22%
R-33	51	343,079	₹305.56 Cr	₹283.95 Cr	₹21.60 Cr	7.07%
R-9 (PUNE)	248	1,872,822	₹6891.45 Cr	₹6406.81 Cr	₹484.65 Cr	7.03%
R-7 (DEHRADUN)	133	2,439,532	₹2106.38 Cr	₹1965.83 Cr	₹140.55 Cr	6.67%
R-12 (PATNA)	125	690,969	₹1156.00 Cr	₹1080.52 Cr	₹75.48 Cr	6.53%

Key Findings

- R-6 (CHENNAI) has the highest deduction rate at 19.13% across 119 hospitals and ₹1129.36 Cr in claims, with ₹216.08 Cr deducted.
- R-1 (NEW DELHI) represents the largest absolute fraud exposure in the database at ₹840.54 Cr deducted from ₹7904.82 Cr claimed (10.63% rate).

PATTERN 2

Priority Audit List

settlement_stat + office_master

TOP HOSPITALS BY TOTAL DEDUCTION AMOUNT

Description: Ranks all empanelled hospitals by total amount deducted across their entire history with ECHS — full database coverage. High total deductions indicate either very high claim volumes or high deduction rates (overbilling); both dimensions are shown below for the top 20.

Table 2.1 — Top 20 Hospitals by Total Deduction

ID	Hospital	State	Claims	Claimed	Deducted	Ded %
367	PARK HOSPITAL (A UNIT OF PARK MEDICENTRES & INST PARK CHAIN	Haryana	123,373	₹751.71 Cr	₹172.54 Cr	22.95%
3149	VIJAY HOSPITAL	Haryana	21,815	₹440.20 Cr	₹149.66 Cr	34.00%
1219	HEALING TOUCH SUPER SPECIALITY HOSPITAL	Haryana	71,465	₹548.75 Cr	₹85.87 Cr	15.65%
60	PARK HOSPITAL - CHOWKHANDI PARK CHAIN	Delhi	23,004	₹290.58 Cr	₹66.80 Cr	22.99%
183	MEDANTA THE MEDICITY - GURGAON	Haryana	341,282	₹564.34 Cr	₹49.12 Cr	8.70%
1161	PARK HOSPITAL(UNIT OF KAILASH SUPERSPECIALITY HO PARK CHAIN	Rajasthan	52,070	₹170.41 Cr	₹48.65 Cr	28.55%
109	GRECIAN SUPER - SPECIALITY HOSPITAL	Chandigarh	50,179	₹332.23 Cr	₹48.05 Cr	14.46%
78	LIVASA HOSP(A UNIT OF IVY HEALTHCARE INFRA PVT L	Chandigarh	202,450	₹579.54 Cr	₹46.96 Cr	8.10%
2856	VIRAT HOSPITAL	Haryana	22,179	₹155.28 Cr	₹46.57 Cr	29.99%
1454	YATHARTH WELLNESS HOSPITAL & TRAUMA CENTRE	Uttar Pradesh	135,915	₹191.39 Cr	₹46.43 Cr	24.26%
2127	PARK HOSPITAL (A UNIT OF PARK MEDICITY INDIA PVT PARK CHAIN	Haryana	20,391	₹201.07 Cr	₹43.29 Cr	21.53%
1042	METRO HOSPITAL & HEART INSTITUTE - NOIDA	Uttar Pradesh	77,696	₹268.39 Cr	₹43.25 Cr	16.11%
399	MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED	Rajasthan	317,616	₹567.76 Cr	₹43.02 Cr	7.58%
519	EMC Super Speciality Hospital Pvt. Ltd.	Punjab	39,376	₹405.04 Cr	₹42.28 Cr	10.44%
34	Fortis Hospital - Mohali	Chandigarh	441,516	₹809.00 Cr	₹41.47 Cr	5.13%
292	Madhuraj Hospital (P) Ltd - Kanpur	Uttar Pradesh	11,406	₹124.12 Cr	₹41.12 Cr	33.13%
514	PARK HOSPITAL (A UNIT OF UMKAL HEALTHCARE PVT LT PARK CHAIN	Haryana	59,211	₹285.23 Cr	₹38.34 Cr	13.44%
17	ARTEMIS MEDICARE SERVICES LIMITED - GURGAON	Haryana	462,187	₹628.60 Cr	₹36.99 Cr	5.88%
1373	PARK HOSPITAL - FARIDABAD PARK CHAIN	Haryana	19,458	₹267.59 Cr	₹36.54 Cr	13.66%
74	INDUS SUPER SPECIALITY HOSPITAL - MOHALI	Chandigarh	99,673	₹332.35 Cr	₹34.07 Cr	10.25%

Key Findings

Park Hospital appears 7 times in the top 20/100 hospitals by deduction — combined across its empanelled units, the chain has had ₹432.87 Cr deducted, more than any other individual hospital or chain identified in the baseline data.

VIJAY HOSPITAL has the highest deduction rate in the top 20 at **34.00%** on ₹440.20 Cr claimed — nearly one-third of everything it bills is rejected by assessors, consistent with systematic overbilling.

PATTERN 3

Independent-Method Corroboration
Hospital-level fraud signals, geo-mapped

CROSS-PATTERN HOSPITAL CONCENTRATION

Description: Each row below is a hospital that was independently flagged by **multiple different fraud-detection analyses** — e.g. a hospital appearing in the revenue-inflation outliers, the emergency-admission misuse list, and the multi-abuse hospital list simultaneously. A hospital flagged by many unrelated detection methods is a far stronger fraud signal than any single pattern alone, because each method tests a different part of the claims lifecycle.

Table 3.1 — Top 20 Hospitals by Number of Independent Patterns Flagging Them

ID	Hospital	State	Independent Signals	Combined Claim Vol.	Combined Exposure
3149	VIJAY HOSPITAL	Haryana	14	170,768	₹2,342.27 Cr
2856	VIRAT HOSPITAL	Haryana	14	119,005	₹894.73 Cr
3019	SAKRA WORLD HOSPITAL	Karnataka	14	61,969	₹85.52 Cr
1197	MOOLCHAND HOSPITAL - Lajpat Nagar - III	Delhi	14	116,366	₹62.92 Cr
2694	STAR HOSPITAL (A UNIT OF OM MEDICENTRE PVT	Rajasthan	13	57,379	₹801.35 Cr
1585	YATHARTH WELLNESS SUPER SPECIALITY HOSPITA	Uttar Pradesh	13	80,491	₹431.44 Cr
1676	METRO HOSPITAL & HEART INSTITUTE - HARIDWA	Uttarakhand	13	17,103	₹158.71 Cr
153	INDIAN SPINAL INJURIES CENTRE - VASANT KUN	Delhi	13	75,974	₹115.76 Cr
1802	MGM NEW BOMBAY HOSPITAL - VASHI	Maharashtra	13	82,463	₹81.55 Cr
514	PARK HOSPITAL (A UNIT OF UMKAL HEALTHCARE	Haryana	12	160,004	₹1,368.68 Cr
274	METRO HEART INSTITUTE - FARIDABAD	Haryana	12	204,729	₹1,292.83 Cr
1454	YATHARTH WELLNESS HOSPITAL & TRAUMA CENTRE	Uttar Pradesh	12	491,481	₹1,064.61 Cr
2826	HUMAN CARE MEDICAL CHARITABLE TRUST	Delhi	12	290,702	₹739.69 Cr
2844	SINGHANIA UNIVERSITY HOSPITAL AND RESEARCH	Rajasthan	12	28,431	₹354.91 Cr
1974	FELIX HOSPITAL - NOIDA	Uttar Pradesh	12	32,495	₹327.32 Cr
1535	ASTER DM HEALTHCARE LIMITED	Kerala	12	118,054	₹323.13 Cr
408	SHRI K M MEMORIAL JAIN HEART & GEN HOSPITA	Rajasthan	12	22,384	₹217.77 Cr
316	Bhargava Hospital - Kanpur	Uttar Pradesh	12	18,051	₹192.95 Cr
3419	MAYO MEDICAL CENTRE PVT LTD	Uttar Pradesh	12	37,008	₹164.80 Cr
1988	PARKASH HOSPITAL - AMRITSAR	Punjab	12	11,346	₹146.81 Cr

Key Findings

VIJAY HOSPITAL (NARNAUL, Haryana) is flagged by **14 independent fraud-detection signals** — the single strongest cross-pattern fraud signal in the database, with a combined flagged exposure of ₹2,342.27 Cr.

Hospitals flagged by 8+ independent patterns should be prioritised for joint audit ahead of hospitals with a high score on any single pattern alone — the convergence of unrelated statistical tests on the same facility sharply reduces the chance of a false positive.

PATTERN 4

CROSS-PATTERN REGIONAL & STATE CLUSTERING

Geographic Rollup of All Flagged Cases
By ECHS region & state

Description: Rolls up the cross-pattern hospital index (Pattern 3) by ECHS region and by state, to show where already-flagged fraud cases concentrate geographically. This is the direct "fraud hotspot map" the Module 9 framework calls for, built from corroborated evidence rather than a single financial metric.

Table 4.1 — ECHS Regions Ranked by Combined Cross-Pattern Exposure

Region	Flagged Hospitals	Distinct Patterns	Combined Claim Vol.	Combined Exposure
R-1 (NEW DELHI)	374	25	7,209,073	₹21,185.21 Cr
R-8 (JAIPUR)	255	25	4,107,474	₹15,290.17 Cr
R-9 (PUNE)	220	20	2,684,101	₹15,212.96 Cr
R-29	290	23	5,380,656	₹12,090.86 Cr
R-4 (BANGALORE)	163	23	3,068,551	₹11,928.20 Cr
R-27	199	22	2,704,925	₹11,092.99 Cr
R-24	189	20	1,748,467	₹7,230.31 Cr
R-30	152	22	2,513,757	₹5,435.83 Cr
R-7 (DEHRADUN)	113	21	3,193,901	₹4,053.93 Cr
R-3 (KOLKATA)	198	21	1,844,573	₹3,942.46 Cr
R-6 (CHENNAI)	103	22	921,677	₹3,740.90 Cr
R-2 (MUMBAI)	100	20	3,977,325	₹3,583.39 Cr
R-17	134	23	1,506,896	₹2,537.42 Cr
R-5 (HYDERABAD)	128	22	1,399,581	₹2,514.24 Cr
R-15	84	22	1,008,673	₹2,425.54 Cr

Table 4.2 — Top 15 States Ranked by Combined Cross-Pattern Exposure

State	Flagged Hospitals	Distinct Patterns	Combined Claim Vol.	Combined Exposure
Haryana	555	23	9,315,878	₹37,544.45 Cr
Punjab	489	22	5,029,419	₹23,355.02 Cr
Uttar Pradesh	558	25	8,242,503	₹20,761.05 Cr
Rajasthan	306	25	3,223,430	₹9,756.09 Cr
Delhi	425	24	3,904,295	₹9,343.84 Cr
Kerala	204	24	6,997,846	₹7,478.05 Cr
Maharashtra	339	24	2,861,754	₹5,699.59 Cr
Chandigarh	53	21	1,572,417	₹5,640.68 Cr
Uttarakhand	114	22	1,701,855	₹3,143.80 Cr

State	Flagged Hospitals	Distinct Patterns	Combined Claim Vol.	Combined Exposure
Andhra Pradesh	148	22	1,709,164	₹2,875.45 Cr
Tamil Nadu	164	20	1,278,399	₹2,684.17 Cr
Karnataka	151	23	1,518,236	₹2,548.30 Cr
West Bengal	80	20	761,647	₹2,196.78 Cr
Bihar	76	21	1,043,343	₹1,837.35 Cr
Nepal	29	17	616,955	₹1,423.37 Cr

Key Findings

- R-1 (NEW DELHI)** has the largest combined cross-pattern exposure (₹21,185.21 Cr across 374 flagged hospitals and 25 distinct fraud patterns) — corroborating the region-level deduction-rate finding in Pattern 1 with independent, claim-level evidence.
- Haryana** leads all states with ₹37,544.45 Cr in combined flagged exposure across 555 hospitals — a geographic concentration that should anchor the next phase of regional field audits.

PATTERN 5

Multi-Unit Brand Concentration
Name-normalised hospital chain rollup

CROSS-PATTERN HOSPITAL CHAIN DETECTION

Description: Groups hospitals by normalised brand name (stripping legal-entity suffixes such as "(A Unit of ... Pvt Ltd)") to detect chains where multiple branches are independently flagged across the fraud framework. The baseline Pattern 2 already showed Park Hospital's 5 units dominating the deduction-amount leaderboard; this table extends that logic using every available pattern, not just settlement_stat deductions.

Table 5.1 — Top 15 Multi-Unit Chains by Combined Cross-Pattern Exposure

Chain	Units	Distinct Patterns	Combined Claim Vol.	Combined Exposure
METRO HEART	9	16	455,053	₹2,969.20 Cr
PARK	5	13	268,470	₹2,277.50 Cr
MAX SUPER	15	15	625,480	₹1,912.28 Cr
PARK GURGAON	2	10	217,650	₹1,894.18 Cr
LIVASA HOSP	2	11	416,135	₹1,861.56 Cr
MEDANTA MEDICITY	2	11	990,315	₹1,842.28 Cr
HEALING TOUCH	4	10	159,894	₹1,774.15 Cr
YATHARTH WELLNESS	4	16	593,034	₹1,499.16 Cr
SRI GURU	6	10	357,108	₹1,300.24 Cr
PARAS HEALTHCARE	2	13	505,533	₹1,211.78 Cr
EMC SUPER	2	10	70,001	₹1,109.59 Cr
IVY HEALTHCARE	2	8	132,861	₹1,072.05 Cr
VENKATESHWAR DWARKA	2	10	420,110	₹1,042.59 Cr
FORTIS MOHALI	2	9	354,165	₹984.49 Cr
AMRITA INSTITUTE	3	9	375,043	₹947.21 Cr

Key Findings

- METRO HEART** is the largest multi-unit cluster by combined exposure, with **9 units** independently flagged across 16 distinct fraud patterns — ₹2,969.20 Cr in combined flagged exposure.
- Chains spreading flagged activity across multiple separately-registered legal entities under one brand (as previously documented for Park Hospital) make per-hospital audit thresholds easier to stay under; any chain appearing here with 3+ units should be audited as a single corporate group, not unit-by-unit.

PATTERN 6

Single-Hospital Funnelling

Referral-syndicate detection

CROSS-REGION REFERRAL USAGE

Description: Reuses the existing "single-hospital funnelling" findings, which already identify polyclinics and referring entities that send the overwhelming majority of their referrals (top_dest_pct) to one specific hospital. Each row is flagged **cross-region** when the polyclinic's city differs from the destination hospital's city — a referring entity routing patients out of its own area to one consistent hospital is a stronger fraud signal than routine local referral.

Table 6.1 — Top 20 Funnelling Pairs by Exposure (Cross-Region Flagged)

Polyclinic	City	Referrals	Top Dest %	Funnels To	Dest. City	Cross-Region	Exposure
Kochi	Kochi	238,623	64.6%	Amrita Institute of Medical Sc	Kochi	NO	₹292.86 Cr
VidyaDhar Nagar	VidyaDhar Nagar	89,748	67.5%	MANIPAL HEALTH ENTERPRISES PRI	JAIPUR	YES	₹262.94 Cr
Ludhiana	Ludhiana	144,948	57.5%	FORTIS HOSPITAL - LUDHIANA	LUDHIANA	NO	₹228.90 Cr
Belgaum	Belgaum	80,533	59.2%	KLES DR. PRABHAKAR KORE HOSPIT	BELAGAVI	YES	₹104.54 Cr
Palampur (RC Yol)	Palampur	112,007	68.1%	VIVEKANAND MEDICAL INSTITUTE	PALAMPUR	NO	₹95.87 Cr
Devlali	Devlali	66,756	63.7%	ASHOKA MEDICOVER HOSPITALS (UN	NASHIK	YES	₹76.45 Cr
Yelahanka	Bangalore	73,575	62.1%	ASTER CMI HOSPITAL (A UNIT OF	BANGALORE	NO	₹60.92 Cr
Kilimanur	Kilimanur	110,214	65%	SREE GOKULAM MEDICAL COLLEGE H	THIRUVANANTHAPURAM	YES	₹51.90 Cr
Vadodara	Vadodara	52,734	67.7%	NARHARI HOSPITAL	VADODARA	NO	₹51.33 Cr
Jagraon	Jagraon	37,702	56.2%	FORTIS HOSPITAL - LUDHIANA	LUDHIANA	YES	₹49.40 Cr
Mainpuri	Mainpuri	17,829	56.3%	PUSHPANJALI HOSPITAL & RESEARC	AGRA	YES	₹48.01 Cr
Udhampur	Udhampur	13,117	92.9%	NARAYANA VAISHNO DEVI SPECIALI	JAMMU & KASHMIR	YES	₹47.55 Cr
Fatehgarh	Fatehgarh	38,041	81%	REGENCY HOSPITAL LTD- KANPUR	KANPUR	YES	₹44.85 Cr
Sholapur	Sholapur	36,128	69.1%	Ashwini Sahakari Rugnalaya Ani	Solapur	YES	₹40.39 Cr
COD Kandivali(Malad)	MUMBAI	48,678	59.6%	WOCKHARDT HOSPITALS LTD.	THANE	YES	₹33.85 Cr
Satna	Satna	33,618	78.1%	MP BIRLA HOSPITAL & PRIYAMVADA	SATNA	NO	₹27.08 Cr
Latur	Latur	39,428	72.9%	Vivekanand Medical Foundation	Latur	NO	₹26.01 Cr

Polyclinic	City	Referrals	Top Dest %	Funnels To	Dest. City	Cross-Region	Exposure
Rajouri	Rajouri	10,669	55.4%	NARAYANA VAISHNO DEVI SPECIALI	JAMMU & KASHMIR	YES	₹25.96 Cr
Panaji (RC Pune)	Panaji	26,712	93%	MANIPAL HOSPITALS - GOA (A UNI	PANAJI	NO	₹22.75 Cr
Tirunelveli	Tirunelveli	34,230	68.8%	Shifa Hospitals	TIRUNELVELI	NO	₹21.79 Cr

Key Findings

- Kochi** (Kochi) is the largest funnelling pair by exposure overall, sending **64.6%** of its 238,623 referrals to a single hospital — **Amrita Institute of Medical Sciences And Research Centre** in Kochi — worth ₹292.86 Cr. This pair is same-city and may reflect ordinary geographic convenience rather than a syndicate.
- 48 of the 78 funnelling pairs identified are cross-region (polyclinic and destination hospital in different cities) — these should be the first priority for referral-syndicate audits, since a same-city referral can still be explained by simple geographic convenience, while a cross-region one cannot. The largest cross-region pair is **VidyaDhar Nagar** (VidyaDhar Nagar) → **MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED** (JAIPUR), worth ₹262.94 Cr.

PATTERN 7

Identify Emerging Clusters
Year-on-year billing acceleration

PREDICTIVE — EMERGING FRAUD CLUSTERS

Description: Reuses year-on-year billing-growth figures already computed elsewhere in the fraud-analytics programme's time-series and specialty-misuse analyses, to flag hospitals whose claim volume or billed amount is accelerating fastest, rather than hospitals that are already large, structural outliers like Park Hospital or Vijay Hospital. A hospital climbing rapidly today is the geographic hotspot of tomorrow — catching it early is materially cheaper than auditing it after the pattern is entrenched.

Table 7.1 — Top 20 Hospitals by Year-on-Year Billing Acceleration (Min. ₹1 Cr Exposure)

ID	Hospital	Source Module	Spike Year	YoY Growth %	Exposure
3149	VIJAY HOSPITAL	M12 Hospital YoY Billing Spike	2024	164%	₹249.71 Cr
2694	STAR HOSPITAL (A UNIT OF OM MEDICENT	M12 Hospital YoY Billing Spike	2025	217%	₹134.24 Cr
2856	VIRAT HOSPITAL	M12 Hospital YoY Billing Spike	2024	623%	₹104.66 Cr
3149	VIJAY HOSPITAL	M12 Hospital YoY Billing Spike	2023	1,117%	₹94.75 Cr
2694	STAR HOSPITAL (A UNIT OF OM MEDICENT	M10 Time-Series Surge (YoY)	2025: 5673	245%	₹76.22 Cr
3201	MAX SUPER SPECIALITY HOSPITAL (UNIT	M12 Hospital YoY Billing Spike	2023	135%	₹62.80 Cr
3153	DR S P YADAV HOSPITAL	M12 Hospital YoY Billing Spike	2025	1,366%	₹61.47 Cr
3232	LAXMI NARAYAN HOSPITAL	M12 Hospital YoY Billing Spike	2024	166%	₹61.05 Cr
274	METRO HEART INSTITUTE - FARIDABAD	M12 Hospital YoY Billing Spike	2023	283%	₹58.99 Cr
34	Fortis Hospital - Mohali	M12 Hospital YoY Billing Spike	2022	105%	₹57.73 Cr
1042	METRO HOSPITAL & HEART INSTITUTE - N	M12 Hospital YoY Billing Spike	2023	236%	₹54.96 Cr
3101	PARAS HEALTHCARE PVT LTD	M12 Hospital YoY Billing Spike	2022	231%	₹53.53 Cr
78	LIVASA HOSP(A UNIT OF IVY HEALTHCARE	M12 Hospital YoY Billing Spike	2021	141%	₹52.43 Cr
2182	THE CORPORATE HOSPITAL	M12 Hospital YoY Billing Spike	2023	107%	₹51.68 Cr
1140	MAX SUPER SPECIALITY HOSPITAL - SHAL	M12 Hospital YoY Billing Spike	2023	122%	₹50.03 Cr
109	GRECIAN SUPER - SPECIALITY HOSPITAL	M12 Hospital YoY Billing Spike	2025	122%	₹49.33 Cr
240	MAX SUPER SPECIALITY HOSPITAL - MOHA	M12 Hospital YoY Billing Spike	2025	5,613%	₹48.71 Cr
1211	SARVODAYA MULTISPECIALITY & CANCE HO	M12 Hospital YoY Billing Spike	2025	102%	₹48.17 Cr
519	EMC Super Speciality Hospital Pvt. L	M12 Hospital YoY Billing Spike	2022	254%	₹47.49 Cr
3101	PARAS HEALTHCARE PVT LTD	M10 Time-Series Surge (YoY)	2022: 1206	257%	₹42.62 Cr

Key Findings

- VIJAY HOSPITAL** shows the sharpest year-on-year acceleration in the dataset — 164% growth reaching ₹249.71 Cr in exposure by 2024.
- Hospitals appearing in both this predictive table and the baseline/cross-pattern hotspot tables (Patterns 1–3) are the highest-confidence audit targets: they are not just structurally high-risk, they are actively getting worse.

CONSOLIDATED SUMMARY

All findings are based on structured database analysis and must be corroborated with physical audit records before enforcement action is taken.

#	Pattern	Coverage	Headline Finding
1	Regional Fraud Hotspots	31 regions	R-6 (CHENNAI) — 19.13% deduction
2	Top Hospitals by Deduction	Top 100 hospitals	Park Hospital chain — ₹432.87 Cr deducted across 7 units
3	Cross-Pattern Hospital Concentration	4,271 hospitals	VIJAY HOSPITAL — 14 independent signals
4	Cross-Pattern Regional & State Clustering	33 regions, 35 states	Haryana — ₹37,544.45 Cr combined exposure
5	Cross-Pattern Hospital Chain Detection	563 multi-unit chains	METRO HEART — 9 units, ₹2,969.20 Cr
6	Cross-Region Referral Usage	78 funnelling pairs (48 cross-region)	VidyaDhar Nagar → MANIPAL HEALTH ENTERPRISES PRIVATE LIMITED (67.5% of referrals)
7	Predictive — Emerging Fraud Clusters	1,066 hospital-year records	VIJAY HOSPITAL — 164% YoY growth

Prepared by IIT Kanpur — Data Analytics & Fraud Intelligence Division | 29 June 2026

All findings are based on structured database analysis and must be corroborated with physical audit records before enforcement action.